Introduction to Artificial Intelligence

CSC 480: Artificial Intelligence, Spring 2026
Professor Austin P. Wright: Course Notes

Abstract
We will cover some of the historical context of the field of artificial intelligence, as well as some provisional initial conceptions of what it means to design intelligent systems and agents.

I What is Artificial Intelligence?

The very first question you should be asking when starting an introduction to some new field of study is: *"What the hell am I getting myself into?"*

This is especially true in a topic as widely discussed, and therefore widely misunderstood, as artificial intelligence. The goal of this course is not to provide you with any once-an-for-all answer to *"What is AI? And how do I build it?"*, as we will see such answers might not be so easy. But perhaps by the end of it you might be able to ask the right questions.

To help us along this journey we will be referencing lots of important and seminal works both historical[1] and contemporary. However our most frequent companion will be the seminal textbook by Stuart Russel and Peter Norvig, *Artificial Intelligence: A Modern Approach* [1]. But do not fear those over 1000(!) pages, the goal of these course notes is to provide a more concise scaffold of the material covered in this course specifically, while gesturing out towards material to check out whenever you want to dive a little bit deeper. That all being said we might as well take a first crack at trying to figure out what we are even talking about with AI.

> ***Artificial Intelligence***
>
> Artificial, from the latin *artificium*, meaning: "A work of art, skill, theory, system, or craft". And Intelligence from the latin *intelligentia* meaning: "understanding, knowledge, comprehension, or the power of discernment". And so perhaps [2] we can proffer our first definition of AI that will guide our journey:
>
> AI is the study of the ***crafting of understanding***.

This is the perspective we will take for the explorations of this course. Our goal being to learn how to design systems for which we can be said to be *crafting* the ways in which they *understand* and *reason*.[3] This emphasis on the thoughtful design of systems of thinking formed the heart of the early days of AI as a field, as well as computer science more generally.

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[1] Well, at least as historical as anything gets in Computer Science

[2] If you will forgive a mild occurrence of the etymological fallacy

[3] If you think there has been some slight-of-hand by already attributing properties like *reason* and *understanding* to machines right from the outset, you are not alone. The philosophical issues underpinning AI as an endeavor are many and deep, and will be discussed later in the course.

II A Brief History of AI

The rest of this document will be laid out much more similarly to how the rest of the course notes for the course will look. I am writing these notes out in order to help keep on-pace and structure my lectures[4], as well as to provide a study resource and citations and links to relevant external material[5]. What I am *not* writing is an entire textbook in real time. Therefore you may expect substantially less detail than what is actually covered in class[6], where you are very much still expected to take your own notes. I hope that these note might help provide the scaffold for your notes in the same way they help scaffold my lecture, and of course if you miss a lecture then these might help point you to the relevant parts of the book or other sources you can use to catch up. But don't expect to be able to just skim through these notes without coming to class, at least if you want to pass.

[4] And of course, since I already know the material what I need in order to stay on top of things is much different than what you need in order to actually learn it.

[5] In particular much of the focus today will be on AIMA chapters 1, 2

[6] And it is the ***material covered in class***, alongside any specifically mentioned readings, that you will be responsible for on the exams.

Thinking about Thinking

Logic and Rationality

- Plato and the reality of concepts
- Aristotle and logic
- Russel, Wittgenstein, and *Logical Positivism*

The Mind

- Cartesian dualism
- Materialism
- Phenomenology
- Searle, The Chinese Room, and *The Hard Problem of Consciousness*
- Behaviorism, Skinner
- Cognitive Psychology

Turing and the Birth of Computing

- Godel Incompleteness
- Computers as Thinking machines
- Turing Test
- Cybernetics, Systems Theory, Feedback, and Interdisciplinarity

> ***Discuss:*** Given the existing history of thought about the nature of consciousness and intelligence at the time of the birth of computing. How would you think about the most important and foundational problems to focus on towards building "Artificial Intelligence"?

Early Optimism

Dartmouth AI Summer Project

- McCarthy, Minsky, Rochester, Shannon, 1955[7]
- First use of term Artificial Intelligence:
- *"2 month, 10 man study ... significant advance in one or more of these problems ... over a summer"*
- McCarthy / Minsky divide: Logical vs Connectionist Systems

Summer Vision Project

- Papert 1966, MIT[8] [9]
  - *"The construction of a significant part of the visual system"*

[7] http://jmc.stanford.edu/articles/dartmouth/dartmouth.pdf

[8] https://dspace.mit.edu/bitstream/handle/1721.1/6125/AIM-100.pdf

[9] https://philippschmitt.com/archive/computer-vision-history/

We propose that a 2 month, 10 man study of artificial intelligence be carried out during the summer of 1956 at Dartmouth College in Hanover, New Hampshire. The study is to proceed on the basis of the conjecture that every aspect of learning or any other feature of intelligence can in principle be so precisely described that a machine can be made to simulate it. An attempt will be made to find how to make machines use language, form abstractions and concepts, solve kinds of problems now reserved for humans, and improve themselves. We think that a significant advance can be made in one or more of these problems if a carefully selected group of scientists work on it together for a summer.

The following are some aspects of the artificial intelligence problem:

1 **Automatic Computers**

If a machine can do a job, then an automatic calculator can be programmed to simulate the machine. The speeds and memory capacities of present computers may be insufficient to simulate many of the higher functions of the human brain, but the major obstacle is not lack of machine capacity, but our inability to write programs taking full advantage of what we have.

2. **How Can a Computer be Programmed to Use a Language**

It may be speculated that a large part of human thought consists of manipulating words according to rules of reasoning and rules of conjecture. From this point of view, forming a generalization consists of admitting a new word and some rules whereby sentences containing it imply and are implied by others. This idea has never been very precisely formulated nor have examples been worked out.

3. **Neuron Nets**

How can a set of (hypothetical) neurons be arranged so as to form concepts. Considerable theoretical and experimental work has been done on this problem by Uttley, Rashevsky and his group, Farley and Clark, Pitts and McCulloch, Minsky, Rochester and Holland, and others. Partial results have been obtained but the problem needs more theoretical work.

4. **Theory of the Size of a Calculation**

If we are given a well-defined problem (one for which it is possible to test mechanically whether or not a proposed answer is a valid answer) one way of solving it is to try all possible answers in order. This method is inefficient, and to exclude it one must have some criterion for efficiency of calculation. Some consideration will show that to get a measure of the efficiency of a calculation it is necessary to have on hand a method of measuring the complexity of calculating devices which in turn can be done if one has a theory of the complexity of functions. Some partial results on this problem have been obtained by Shannon, and also by McCarthy.

5. **Self-Improvement**

Probably a truly intelligent machine will carry out activities which may best be described as self-improvement. Some schemes for doing this have been proposed and are worth further study. It seems likely that this question can be studied abstractly as well.

6. **Abstractions**

A number of types of "abstraction" can be distinctly defined and several others less distinctly. A direct attempt to classify these and to describe machine methods of forming abstractions from sensory and other data would seem worthwhile.

7. **Randomness and Creativity**

A fairly attractive and yet clearly incomplete conjecture is that the difference between creative thinking and unimaginative competent thinking lies in the injection of a some randomness. The randomness must be guided by intuition to be efficient. In other words, the educated guess or the hunch include controlled randomness in otherwise orderly thinking.

Figure 1: First page from Proposal for Summer Research Project in Artificial Intelligence. AI involves the corse of every other part of computer science.[10]

[10] *AI is just the parts of CS we have yet to fully understand*

1. Automatic Computers ⇒ Hardware / Computer Architecture
2. How Can a Computer be Programmed to Use a Language ⇒ PL (LISP : McCarthy 1958) / Formal Methods / NLP
3. Neuron Nets ⇒ Deep Learning
4. Theory of the Size of a Calculation ⇒ Complexity Theory
5. Self Improvement ⇒ Machine Learning
6. Abstractions ⇒ Knowledge Representation, OOP/Software Engineering
7. Randomness and Creativity ⇒ Probabilistic and Stochastic Computing

The First AI Winter

- Boom/Bust cycle, the runaway effect of the popular imagination
- Lack of formal treatment of the difficulty of the problems being attempted
- Scale and hardware limitations
- Perceptrons, Minsky, XOR, No Multilayer Backpropagation
- 1974 Lighthill Report

Rise of Expert Systems

- Attempts to solve more *well defined and limited* problem with use of *expert background knowledge*
- Knowledge expressible as *rules*
  - > ***if*** M is the mass of the whole molecule ***and*** there are two peaks at x_1 and x_2 such that
    - $x_1 + x_2 = M + 28$
    - $x_1 - 28$ is a high peak
    - $x_2 - 28$ is a high peak
    - At least one of x_1 and x_2 is high
    >
    > ***then*** there is a ketone subgroup.
- LISP, Symbolic Logic
- Prolog, Logic Programming,
- Second AI Winter

Games

- Chess, Deep Blue vs Gary Kasparov, 1996
- Jeopardy, IBM Watson vs Ken Jennings, 2011
- Go, Alphago vs Lee Sedol, 2016

> ***Discuss:*** Think of some other games, be it video games, board games, sports, or something else. Is AI already better than humans? If so, when and using what methods? If not, why, and what are the things preventing it?

Big Data and Machine Learning

> "Every time I fire a linguist, the performance of the speech recognizer goes up" - Frederick Jelinek

- Many hard real world problems are hard to explicitly model effectively
- Richard Sutton, The Bitter Lesson [11]
- Unreasonable Effectiveness of Data [12]
- From Machine learning into Deep Learning

Large Language Models and the Current Frontier

- Language as a general purpose environment
- Essential properties of *scale*
- Return to the early optimism of general thinking things
- Have we lost what we were trying to do? Or have we finally caught up?

[11] https://www.cs.utexas.edu/~eunsol/courses/data/bitter_lesson.pdf

[12] https://ieeexplore.ieee.org/document/4804817

III Intelligent Agents

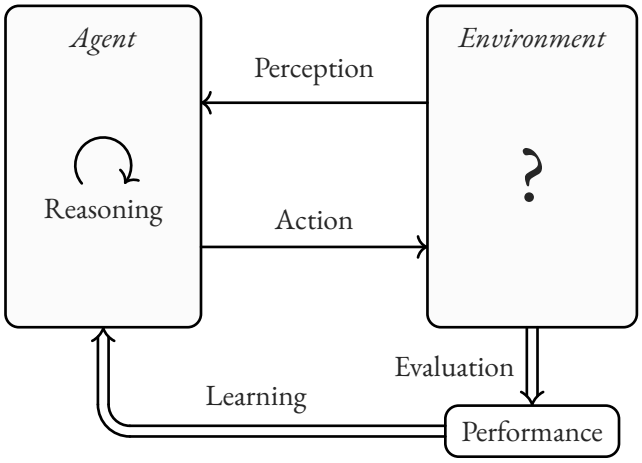

Figure 2: Basic Components of Agents

> ***Definition: Rational Agent***
>
> A rational agent is an agent for which, for each possible percept sequence, it should select an action that is expected to maximize its performance measure, given the evidence provided by the percept sequence and whatever built-in knowledge the agent has.

Kinds of Environment

- Static vs Dynamic
- Fully vs Partially Observable
- Deterministic vs Stochastic
- Discrete vs Continuous
- Single vs Multi-Agent

Kinds of Agent

- Reflex Agent
- Model Based Agent
  - Atomic, Factored, or Structured State Spaces
- Goal Based Agent
- Utility Based Agent
- Learning Agent

> ***Discuss:*** Compare two different "AI" agents, ChatGPT[13] and Stockfish (current best chess bot). Using the agentic framework, how are these agents the same? different? In which ways do either exhibit essential characteristics of "intelligence"? What about "rationality"? Are there any other interesting examples you can think of?

[13] Get ready that this is basically the last of any LLM content in the course!

IV Exercises

Course Intro Lab Activity

1. Enroll in the course sites (EdStem and Gradescope)
2. Post a short introduction in the "Get to know your class" thread on EdStem
3. Complete the following exercise to show you understand the course grading policy, and test it out on plenty of possible outcomes!

Write a short python function that takes a list of exam scores, a list of lab scores, and a final project score and outputs a final letter grade in accordance with the Syllabus grading policy.

```
def grader(exam_scores, lab_scores, final_project):
    return final_grade
```

Bibliography

[1] S. Russell and P. Norvig, *Artificial Intelligence: A Modern Approach, 4th US ed*. Pearson, 2022. [Online]. Available: https://aima.cs.berkeley.edu/